NEXUM SUUM
Incorporated

✅ SOP #2 – ATI (Analysis to Improve) for Boilers

Applies to: All operating boilers used for facility or process heating
Objective: Improve efficiency, performance, and compliance via analysis of inputs/outputs, trends, and anomalies.

📊 1. Performance Metrics to Monitor

Metric	Ideal Target	Optimization Tip
Combustion Efficiency	>82% (non-condensing); >90% condensing	Tune annually, verify O_2 & CO levels
Fuel-to-Steam Ratio	1 therm = ~100,000 BTU steam	Reduce excess air, check burner settings
Stack Temperature	Minimize to avoid energy loss	Add or clean economizer
Cycles of Concentration	5–10 for steam systems	Adjust blowdown based on conductivity
Downtime between runs	<2 hours	Use lead-lag logic or modulating controls

🔍 2. Key Improvement Actions

Install O_2 Trim Controls: Reduces excess air based on real-time combustion analysis
Optimize Blowdown: Reduce waste and conserve treated water
Upgrade to VFD Feedwater Pumps: Avoid over-pumping during low demand
Install Insulation Wraps: On exposed valves, headers, PRVs
Data Logging & Pattern Alerts: Use dashboard to set alarms on efficiency drop or cycling

⚖️ 3. Legal, Ethical & Safety Notes

ASME Code Compliance: All changes must meet BPVC Section I or IV
OSHA 1910.109 Regulation: On flammable fuels and pressure vessels
EPA Compliance: Tune-ups and combustion logs required for many large units
Insurance Audits: Required logs should be maintained for liability reviews